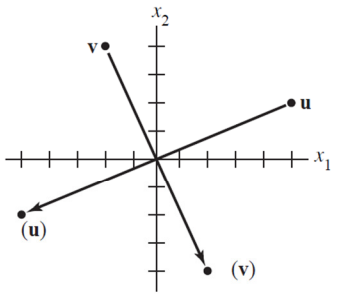
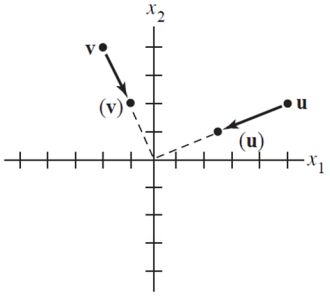
1.8 SOLUTIONS

13.



A reflection through the origin.

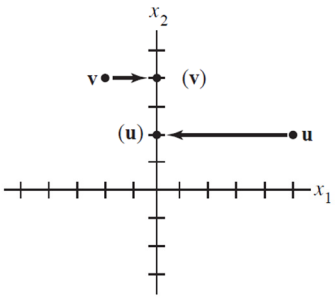
14.



A contraction by the factor .5.

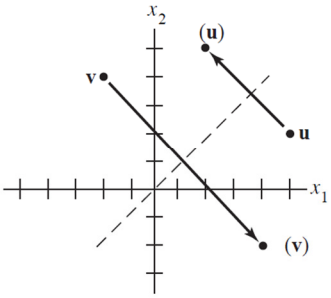
The transformation in Exercise 13 may also be described as a rotation of π radians about the origin or a rotation of $-\pi$ radians about the origin.

15.



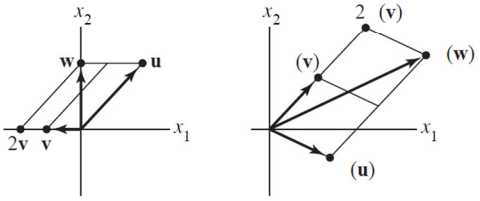
A projection onto the x_2-axis

16.



A reflection through the line $x_2 = x_1$.

18. Draw a line through **w** parallel to **v**, and draw a line through **w** parallel to **u**. See the left part of the figure below. From this, estimate that $\mathbf{w} = \mathbf{u} + 2\mathbf{v}$. Since T is linear, $T(\mathbf{w}) = T(\mathbf{u}) + 2T(\mathbf{v})$. Locate $T(\mathbf{u})$ and $2T(\mathbf{v})$ as in the right part of the figure and form the associated parallelogram to locate $T(\mathbf{w})$.



20. Use the basic definition of $A\mathbf{x}$ to construct A. Write

$$T(\mathbf{x}) = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 = \begin{bmatrix}\mathbf{v}_1 & \mathbf{v}_2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} = \begin{bmatrix}-2 & 7\\ 5 & -3\end{bmatrix}\mathbf{x}, \quad A = \begin{bmatrix}-2 & 7\\ 5 & -3\end{bmatrix}$$